linked list

Linked list

- The sequence of data structures that are connected with a link is called a linked list.
- Link: Each link of the linked list can store the data called an element.
- The linked list is a linear data structure.
- Next: Each link of the linked list contains the link to the next link called next.

Types of linked list

Simple linked list: item navigation is forward only.

Doubly linked list: item navigation is both forward and backward.

Circular linked list: last item contains link of the first element as next.

Basic operation:

The basic operations performed in a linked list is

Insertion: adds a new element at the beginning of the list.

Deletion: deletes an element at the beginning of the list.

Display: displays the complete list.

Search: searches of the targeted item in the list.

Delete, delete any targeted item from the list.

Representation of the linked list

There are 2 types of representation

- Static representation.(using arrays)
- Dynamic representation.(using free pool of memory)

Algorithms

Traversing a linked

list, its a process of visiting all the elements or nodes in the linked list exactly once algorithm for traversing linked list

Step1: start

Step 2:intializing PTR= start

Step 3: while(PTR!= NULL) do

Process PTR->info

Set PTR =PTR->link

End of while

Step 5: stop

Searching an element in a linked list

Step 1: start

Step 2: initialize PTR= start, LOC=NULL

Step 3: while(PTR!= NULL) do

if(PTR->info==item)

LOC=PTR

Go to step 4

Else PTR=PTR->link

End of if

End of while

Step 4: if(LOC== NULL)

Print search unsuccessful

Else

Print: search successful and found at LOC

End of if

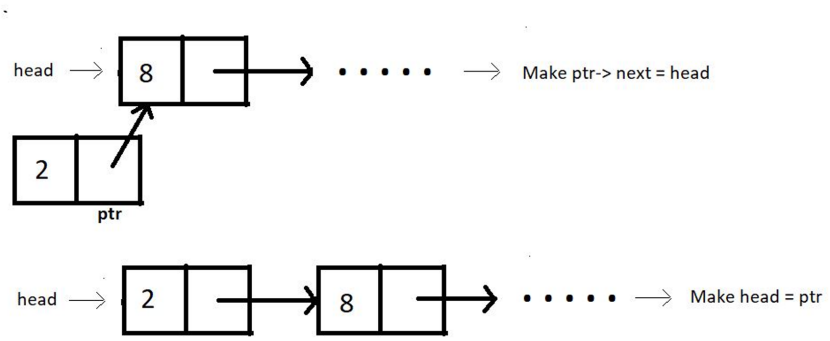
Step 5: stop

insertion of a node in a linked list

insertion in a linked list can be performed in a few different ways it's the insertion of a new node in the linked list. the .types of a linked list are:

- insertion in the beginning
- insertion in the end
- insertion after a specific value
- insertion in a sorted linked list

insertion in the beginning



step 1: start

step 2: if AVAIL= NULL

then print: overflow

step 3: set NEW=AVAIL

AVAIL=AVAIL→link

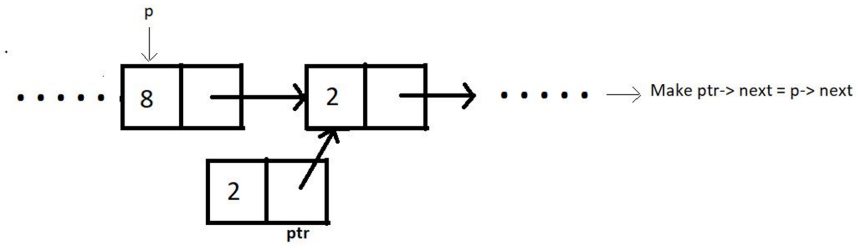
step 4: set NEW→ INFO= item

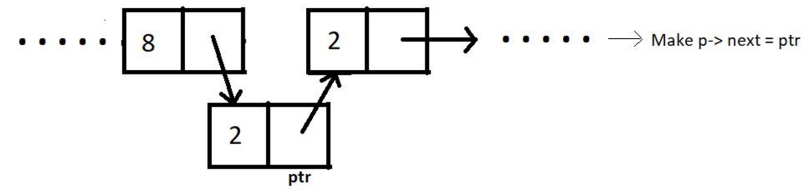
step 5: set NEW→link= START

step 6: set START= NEW

step 7: stop

insertion of an element at the end of the list





step 1: start

step 2: if AVAIL= NULL

then print: overflow

step 3: set NEW=AVAIL

AVAIL=AVAIL→link

step 4: set NEW→ INFO= item

step 5: if START=NULL then

START = NEW

NEW→link=NULL

ELSE

PTR=START

while(PTR≠NULL)

PTR=PTR→link

end of while

PTR→link=NEW

NEW→link=NULL

end of if

step 6: stop

insertion of a node in a specific location

step 1: start

step 2: if AVAIL=NULL then

print: overflow

go to step 6

step 3: set NEW=AVAIL

AVAIL=AVAIL→link

step 4: set NEW→info=item

set NEW→link=NULL

step 5:LOC=1 then

NEW→link=START

START=NEW

ELSE

set PTR=START, COUNT=1

while(COUNT! =LOC-1)

PTR=PTR→link

count++

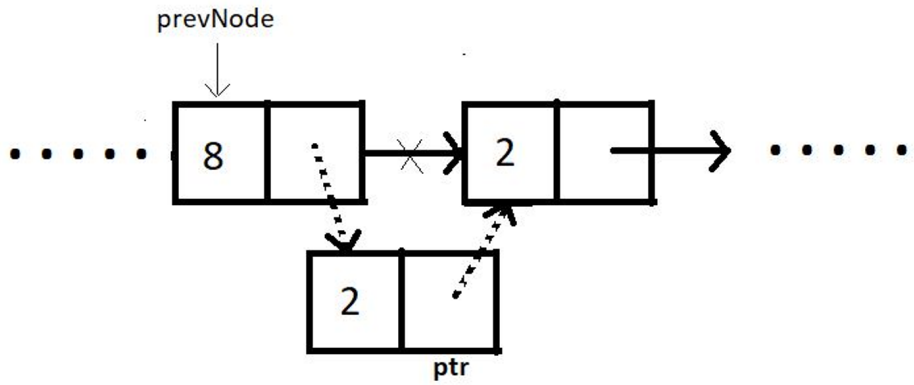
End of while

NEW→link=PTR→link

PTR→link=NEW

step 6:stop

insertion of a node after a specific node or value



step 1: start

step 2: